Initial Solution V2.1
Cache Aware Recourse Report

One slot reserve parcel, paired experiment এবং ফলাফল

Project: PGBM thesis and simulation work
সমাধানের ধরন: Cache aware one slot recourse
Environment: Synthetic এবং DU outdoor
মোট configuration: ২৪০টি paired configuration, ৭২০টি metrics row
Seed: ১০১–১১০
Report date: ৭ অক্টোবর ২০২৬

এই report-এ V2.1 cache loading rule, feasibility guard, paired V2 control, cache hit evidence এবং one slot policy decision একই flow-তে ব্যাখ্যা করা হয়েছে।

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1 V2.1-এর core feature

V2.1 হলো V2-এর উপর নির্মিত bounded cache aware recourse layer। Ordinary V2 mission route তৈরি হওয়ার পরে dispatch time-এ সর্বোচ্চ একটি reserve parcel load করা হয়। Payload slot, payload mass, full route energy, reserve energy, return route এবং ১২০ মিনিটের horizon feasible হলেই reserve load হয়। Reserve future undetected task দেখে নির্বাচন করা হয় না এবং flight-এর মাঝখানে reload করা হয় না। পরবর্তী replacement V2-এর one-for-one positive service-value gain rule অনুসরণ করে।

- At most one reserve parcel: একটি mission-এ সর্বোচ্চ একটি cache slot।
- Feasibility guards: payload, mass, energy reserve, return এবং horizon।
- Configured prior: default uniform prior এবং labelled water-heavy sensitivity prior।
- Cache evidence: load, use, match, accepted hit, miss এবং returned parcel আলাদা record।
- V2 isolation: canonical V1 এবং V2 code path ও evidence overwrite করা হয়নি।

2 Experiment-এর সংক্ষিপ্ত চিত্র

V2 control এবং প্রতিটি V2.1 policy একই seeded scenario-তে চালানো হয়েছে। দুইটি environment, ৩০/৬০/৯০ task load, ৩/৫/৮/১০ UAV এবং ১০টি seed ব্যবহার করে ২৪০টি scenario, ৭২০টি metrics row এবং ২৪০টি decision trace তৈরি হয়েছে। Matrix generation-এর জন্য bounded planner settings manifest-এ সংরক্ষিত আছে।

3 Cache loading এবং recourse flow

প্রথমে ordinary V2 route calculate হয়। এরপর configured prior থেকে একটি item নির্বাচন করা হয়। Free payload slot, remaining mass, cache সহ full route energy, reserve margin, return time এবং horizon check করা হয়। সব guard pass করলে reserve inventory mission state-এ রাখা হয়। Later replacement-এ cache বা released displaced inventory candidate demand satisfy করতে পারে, কিন্তু candidate শুধুমাত্র feasible positive service-value gain হলে accepted hit হয়। Loaded কিন্তু unused parcel mission শেষে returned হিসেবে record হয়।

4 Aggregate performance

সারণী 1: V2 control এবং V2.1 policy aggregate result।

Variant	Episodes	Objective	Served	Delivery	Energy	Delay
V2 control	240	17.01 ± 9.36	27.39 ± 15.40	48.2%	5613.94 ± 2225.71	18.23 ± 4.68
Uniform cache	240	17.25 ± 9.45	27.68 ± 15.65	49.0%	5679.48 ± 2244.19	17.86 ± 4.54
Water heavy sensitivity	240	17.34 ± 9.44	27.84 ± 15.52	49.2%	5704.77 ± 2264.83	18.07 ± 4.55

5 Paired core findings

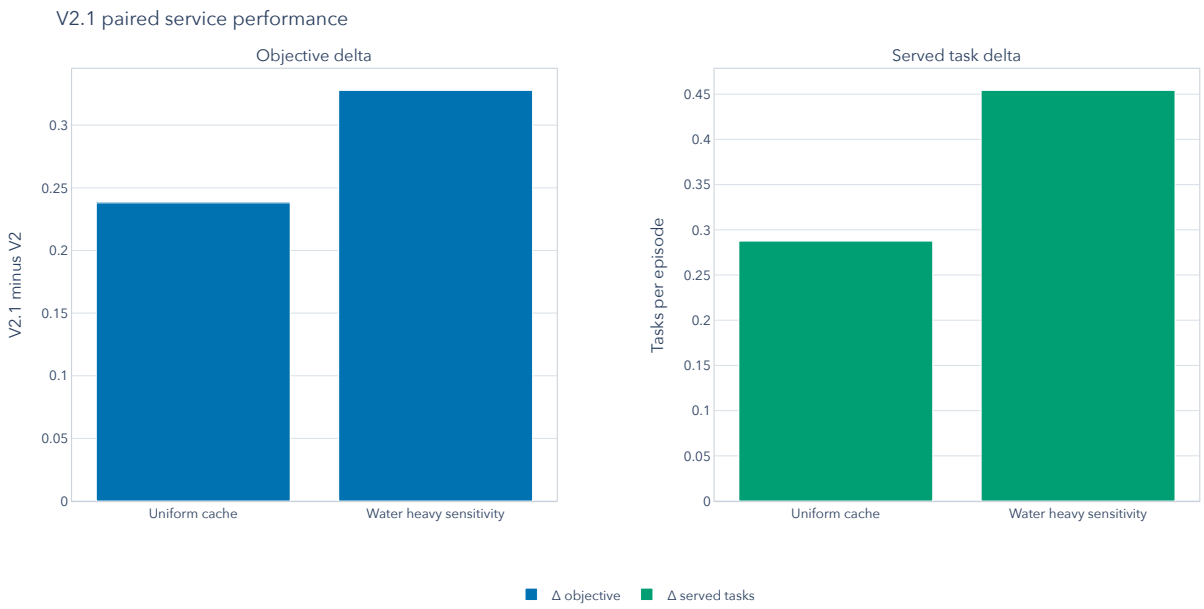
প্রতিটি V2.1 row একই environment, task load, UAV count এবং seed-এর V2 control row-এর সঙ্গে তুলনা করা হয়েছে। দুইটি one-slot policy-তেই objective সামান্য বেড়েছে। Uniform policy-তে mean objective gain 0.238 এবং served task

সারণী 2: V2.1 minus V2 paired delta।

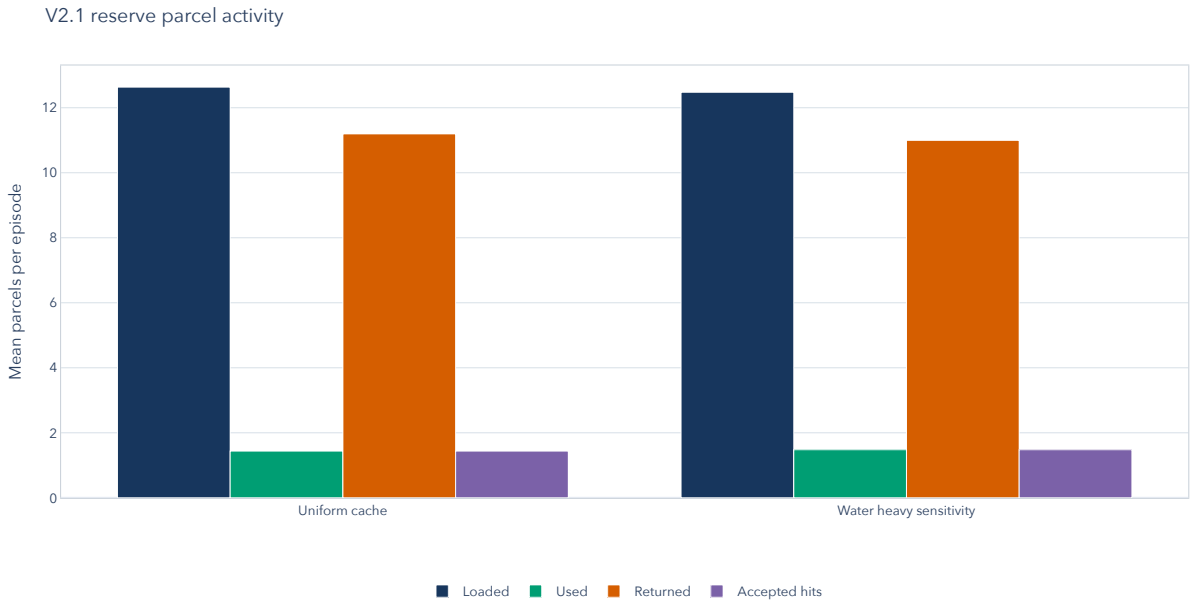
Variant	Episodes	Δ Objective	Δ Served	Δ Deferred	Δ Delay	Δ Energy
Uniform cache	240	0.238	0.287	-0.287	-0.376	65.541
Water heavy sensitivity	240	0.328	0.454	-0.454	-0.163	90.831

gain 0.288। Water-heavy sensitivity-তে gain 0.328 এবং 0.454। তবে reserve load-এর বড় অংশ unused অবস্থায় return করেছে এবং water-heavy policy-তে cache energy overhead বেশি।

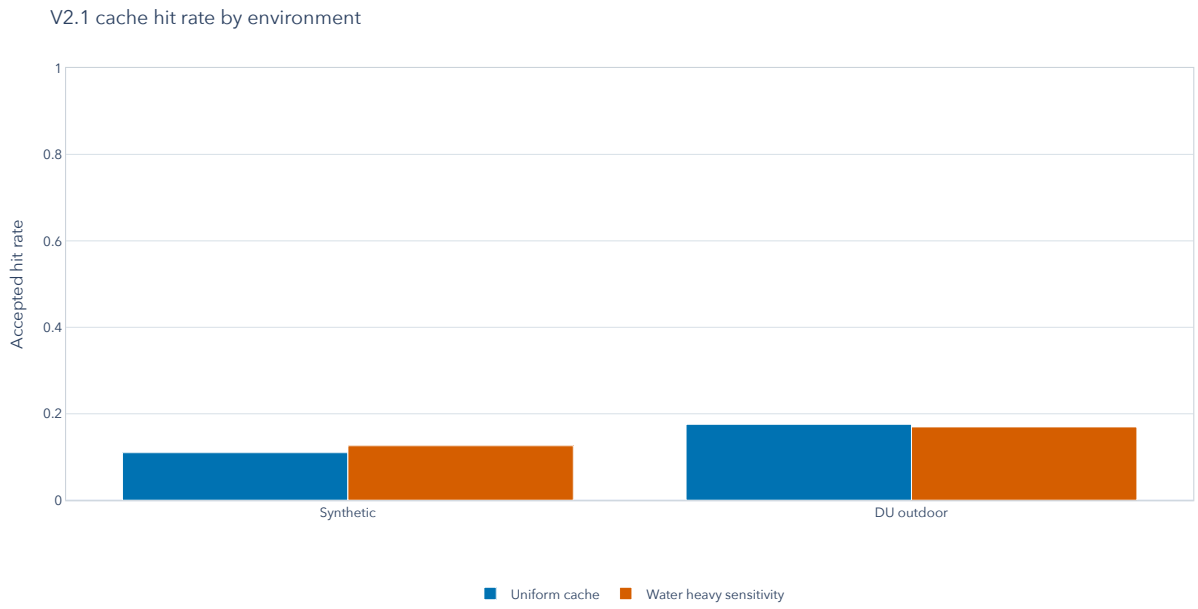
6 Cache activity এবং environment sensitivity



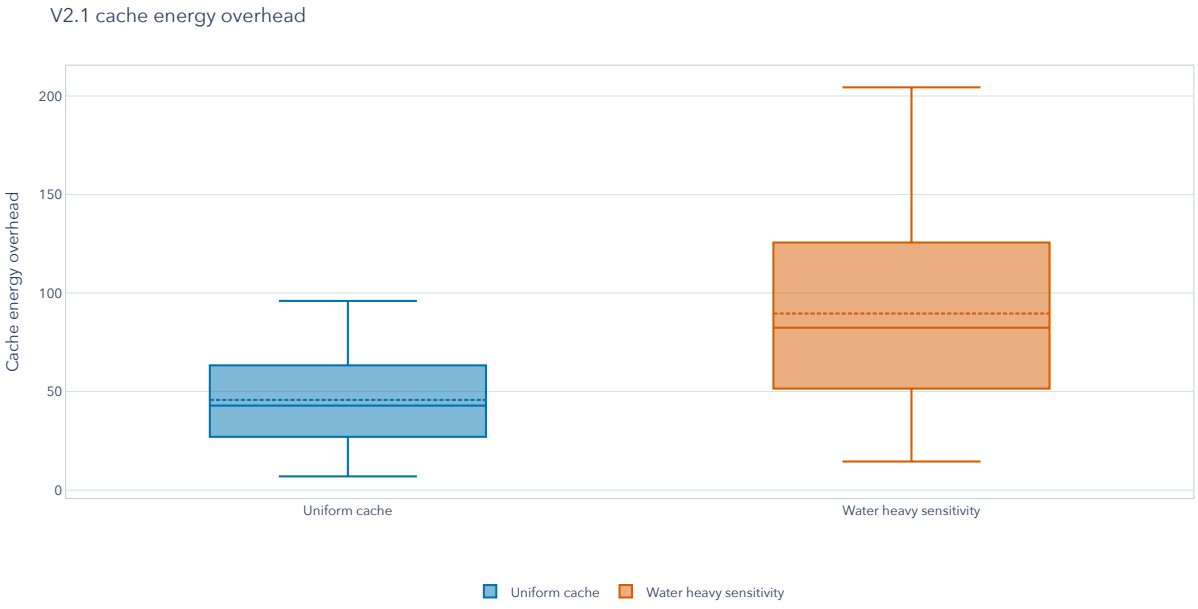
চিত্র 1: V2.1 paired objective এবং served task delta।



চিত্র 2: Reserve parcel load, use, return এবং accepted hit।



চিত্র 3: Environment অনুযায়ী cache hit rate।



চিত্র 4: Cache energy overhead distribution।

সারণী 3: Environment, task load, fleet এবং policy অনুযায়ী সম্পূর্ণ V2.1 matrix summary।

Environment	Task	UAV	Variant	Objective	Served	Delivery	Energy	Hit rate	Overhead
DU outdoor	30	3	V2 control	5.54 ± 1.11	9.10 ± 2.33	25.3%	2863.48 ± 223.88	0.0%	0.00 ± 0.00
DU outdoor	30	3	Uniform cache	5.73 ± 1.13	9.00 ± 2.00	26.0%	2922.77 ± 204.75	19.2%	25.33 ± 5.30
DU outdoor	30	3	Water heavy sensitivity	5.89 ± 1.05	9.70 ± 2.00	27.0%	2961.20 ± 257.12	12.1%	53.81 ± 16.16
DU outdoor	30	5	V2 control	7.80 ± 1.42	13.80 ± 3.36	39.9%	4616.14 ± 312.52	0.0%	0.00 ± 0.00
DU outdoor	30	5	Uniform cache	8.29 ± 1.44	14.30 ± 3.20	41.4%	4673.24 ± 281.94	17.0%	41.82 ± 9.83
DU outdoor	30	5	Water heavy sensitivity	8.34 ± 1.55	14.50 ± 3.54	43.4%	4696.47 ± 328.31	17.5%	80.49 ± 22.33
DU outdoor	30	8	V2 control	10.15 ± 1.69	19.20 ± 3.49	59.7%	6775.10 ± 391.17	0.0%	0.00 ± 0.00
DU outdoor	30	8	Uniform cache	10.41 ± 1.61	19.60 ± 3.72	61.4%	6808.41 ± 455.13	31.0%	64.59 ± 6.76
DU outdoor	30	8	Water heavy sensitivity	10.18 ± 1.78	19.10 ± 3.90	59.1%	6768.38 ± 463.68	16.1%	130.35 ± 16.61
DU outdoor	30	10	V2 control	10.85 ± 1.53	20.80 ± 3.12	65.7%	7700.52 ± 574.49	0.0%	0.00 ± 0.00
DU outdoor	30	10	Uniform cache	10.95 ± 1.50	20.80 ± 3.16	66.1%	7755.53 ± 572.91	16.8%	81.87 ± 8.83
DU outdoor	30	10	Water heavy sensitivity	10.86 ± 1.52	20.70 ± 3.02	65.3%	7787.81 ± 566.99	20.8%	162.59 ± 17.26
DU outdoor	60	3	V2 control	7.01 ± 1.19	10.10 ± 1.66	13.6%	3027.15 ± 216.04	0.0%	0.00 ± 0.00
DU outdoor	60	3	Uniform cache	6.86 ± 0.97	9.80 ± 1.32	13.8%	3016.09 ± 270.92	12.9%	26.15 ± 4.54
DU outdoor	60	3	Water heavy sensitivity	7.01 ± 1.03	10.00 ± 1.49	13.6%	3070.78 ± 194.24	14.5%	52.00 ± 9.95
DU outdoor	60	5	V2 control	9.85 ± 1.41	14.90 ± 1.97	20.7%	4886.49 ± 234.63	0.0%	0.00 ± 0.00
DU outdoor	60	5	Uniform cache	10.03 ± 1.51	15.10 ± 2.08	22.6%	5002.73 ± 188.73	18.3%	45.57 ± 8.91
DU outdoor	60	5	Water heavy sensitivity	9.76 ± 1.65	14.60 ± 2.17	21.8%	4744.61 ± 301.42	14.6%	76.92 ± 17.43
DU outdoor	60	8	V2 control	13.79 ± 1.87	22.40 ± 2.59	34.3%	7548.30 ± 510.66	0.0%	0.00 ± 0.00
DU outdoor	60	8	Uniform cache	13.56 ± 2.14	21.90 ± 3.00	34.2%	7533.59 ± 343.52	15.7%	67.95 ± 9.99
DU outdoor	60	8	Water heavy sensitivity	14.40 ± 2.38	23.30 ± 3.53	36.3%	7789.55 ± 383.07	15.0%	128.25 ± 28.14
DU outdoor	60	10	V2 control	15.85 ± 2.80	26.50 ± 4.48	41.8%	9270.10 ± 512.47	0.0%	0.00 ± 0.00
DU outdoor	60	10	Uniform cache	15.77 ± 2.79	26.50 ± 4.30	42.4%	9210.35 ± 627.87	18.6%	80.46 ± 9.47
DU outdoor	60	10	Water heavy sensitivity	16.56 ± 2.48	27.90 ± 4.09	44.8%	9516.09 ± 673.24	19.4%	159.36 ± 21.68
DU outdoor	90	3	V2 control	6.77 ± 1.50	9.40 ± 1.96	8.4%	2962.64 ± 230.76	0.0%	0.00 ± 0.00
DU outdoor	90	3	Uniform cache	7.23 ± 1.42	9.90 ± 1.79	9.1%	3125.02 ± 222.61	13.2%	27.36 ± 4.71
DU outdoor	90	3	Water heavy sensitivity	6.93 ± 1.36	9.60 ± 1.65	9.3%	2965.24 ± 282.12	25.0%	54.76 ± 8.40
DU outdoor	90	5	V2 control	10.39 ± 2.50	14.80 ± 3.12	14.0%	4748.06 ± 275.69	0.0%	0.00 ± 0.00
DU outdoor	90	5	Uniform cache	10.76 ± 2.21	15.00 ± 2.67	15.0%	4860.48 ± 232.78	12.5%	44.37 ± 7.75
DU outdoor	90	5	Water heavy sensitivity	11.07 ± 2.05	15.90 ± 2.18	15.8%	4905.60 ± 237.10	15.2%	83.34 ± 21.91
DU outdoor	90	8	V2 control	15.10 ± 2.94	22.60 ± 3.69	22.6%	7620.26 ± 381.34	0.0%	0.00 ± 0.00
DU outdoor	90	8	Uniform cache	15.73 ± 2.44	23.30 ± 3.27	24.0%	7795.50 ± 457.83	17.4%	69.94 ± 8.54
DU outdoor	90	8	Water heavy sensitivity	15.85 ± 3.12	23.50 ± 3.95	24.2%	7738.56 ± 351.28	15.0%	131.39 ± 25.03
DU outdoor	90	10	V2 control	17.90 ± 3.14	27.40 ± 4.38	28.3%	9396.49 ± 530.71	0.0%	0.00 ± 0.00
DU outdoor	90	10	Uniform cache	18.78 ± 2.99	28.90 ± 4.23	30.9%	9614.24 ± 388.53	18.0%	83.83 ± 10.25
DU outdoor	90	10	Water heavy sensitivity	18.38 ± 3.15	28.10 ± 4.53	29.4%	9379.10 ± 513.60	18.4%	170.95 ± 28.77
Synthetic	30	3	V2 control	12.26 ± 1.35	21.00 ± 2.16	65.5%	2749.05 ± 186.38	0.0%	0.00 ± 0.00
Synthetic	30	3	Uniform cache	12.38 ± 1.29	21.20 ± 2.04	66.2%	2763.38 ± 187.52	7.4%	14.05 ± 4.28
Synthetic	30	3	Water heavy sensitivity	12.47 ± 1.35	21.10 ± 2.18	66.3%	2751.32 ± 189.08	13.6%	25.71 ± 7.74
Synthetic	30	5	V2 control	15.30 ± 0.98	27.00 ± 1.56	89.0%	4080.53 ± 284.09	0.0%	0.00 ± 0.00
Synthetic	30	5	Uniform cache	15.16 ± 0.98	26.70 ± 1.34	87.3%	4093.29 ± 273.96	8.6%	33.29 ± 5.99
Synthetic	30	5	Water heavy sensitivity	15.37 ± 0.98	27.10 ± 1.45	89.4%	4164.14 ± 324.09	12.9%	67.24 ± 14.75
Synthetic	30	8	V2 control	16.28 ± 1.11	27.50 ± 1.43	90.9%	5003.03 ± 490.94	0.0%	0.00 ± 0.00
Synthetic	30	8	Uniform cache	16.28 ± 1.11	27.50 ± 1.43	90.9%	5061.38 ± 526.72	23.3%	56.33 ± 8.49
Synthetic	30	8	Water heavy sensitivity	16.29 ± 1.09	27.50 ± 1.43	90.9%	5114.90 ± 507.46	5.0%	113.07 ± 16.67
Synthetic	30	10	V2 control	16.44 ± 1.12	27.50 ± 1.43	90.9%	5209.87 ± 552.00	0.0%	0.00 ± 0.00
Synthetic	30	10	Uniform cache	16.44 ± 1.12	27.50 ± 1.43	90.9%	5271.93 ± 559.37	0.0%	62.06 ± 7.44
Synthetic	30	10	Water heavy sensitivity	16.44 ± 1.12	27.50 ± 1.43	90.9%	5333.99 ± 566.75	0.0%	124.12 ± 14.88
Synthetic	60	3	V2 control	15.54 ± 1.62	22.60 ± 2.72	33.5%	2819.10 ± 133.40	0.0%	0.00 ± 0.00
Synthetic	60	3	Uniform cache	15.44 ± 1.65	22.10 ± 2.64	33.0%	2816.79 ± 88.98	10.6%	15.68 ± 6.79
Synthetic	60	3	Water heavy sensitivity	15.50 ± 1.30	22.50 ± 2.07	33.9%	2863.78 ± 126.32	10.4%	29.38 ± 8.76
Synthetic	60	5	V2 control	22.47 ± 1.92	34.90 ± 3.98	54.6%	4547.61 ± 225.19	0.0%	0.00 ± 0.00
Synthetic	60	5	Uniform cache	22.46 ± 1.97	34.40 ± 3.78	53.7%	4534.45 ± 193.67	9.3%	25.37 ± 2.10
Synthetic	60	5	Water heavy sensitivity	22.61 ± 1.83	34.70 ± 3.27	54.4%	4639.60 ± 214.22	16.6%	52.53 ± 11.51
Synthetic	60	8	V2 control	29.51 ± 2.47	50.20 ± 5.55	81.6%	7001.95 ± 249.08	0.0%	0.00 ± 0.00
Synthetic	60	8	Uniform cache	29.78 ± 2.49	50.50 ± 4.74	82.1%	7002.92 ± 344.91	15.7%	41.45 ± 8.59

Environment	Task	UAV	Variant	Objective	Served	Delivery	Energy	Hit rate	Overhead
Synthetic	60	8	Water heavy sensitivity	29.85 ± 2.37	50.90 ± 4.65	83.0%	7127.86 ± 338.59	15.9%	82.12 ± 13.58
Synthetic	60	10	V2 control	31.67 ± 1.87	54.30 ± 2.83	89.4%	8068.17 ± 540.13	0.0%	0.00 ± 0.00
Synthetic	60	10	Uniform cache	31.80 ± 1.69	54.60 ± 2.32	90.1%	8169.13 ± 583.65	10.6%	61.12 ± 8.83
Synthetic	60	10	Water heavy sensitivity	31.87 ± 1.65	54.80 ± 1.99	90.4%	8306.50 ± 653.75	20.3%	124.17 ± 18.99
Synthetic	90	3	V2 control	16.61 ± 1.60	22.70 ± 2.16	21.8%	2814.47 ± 115.30	0.0%	0.00 ± 0.00
Synthetic	90	3	Uniform cache	17.10 ± 1.49	23.10 ± 1.91	22.8%	2885.27 ± 92.69	10.5%	15.59 ± 3.71
Synthetic	90	3	Water heavy sensitivity	17.41 ± 1.60	23.80 ± 2.25	23.1%	2903.20 ± 92.92	14.4%	25.01 ± 6.83
Synthetic	90	5	V2 control	25.36 ± 2.45	36.50 ± 3.66	35.8%	4699.38 ± 157.17	0.0%	0.00 ± 0.00
Synthetic	90	5	Uniform cache	25.77 ± 2.14	37.30 ± 3.13	37.4%	4803.85 ± 166.39	10.1%	24.53 ± 6.87
Synthetic	90	5	Water heavy sensitivity	26.04 ± 2.15	37.40 ± 3.24	37.0%	4808.84 ± 98.43	13.1%	46.33 ± 10.29
Synthetic	90	8	V2 control	35.30 ± 3.12	55.30 ± 5.38	57.7%	7364.18 ± 246.65	0.0%	0.00 ± 0.00
Synthetic	90	8	Uniform cache	35.97 ± 3.06	56.40 ± 4.93	59.2%	7426.48 ± 216.50	11.6%	41.14 ± 8.13
Synthetic	90	8	Water heavy sensitivity	35.72 ± 2.97	55.60 ± 4.97	58.5%	7453.48 ± 230.75	14.9%	76.37 ± 18.88
Synthetic	90	10	V2 control	40.55 ± 3.31	66.80 ± 6.00	71.2%	8962.54 ± 239.81	0.0%	0.00 ± 0.00
Synthetic	90	10	Uniform cache	41.29 ± 2.93	68.80 ± 4.59	74.3%	9160.76 ± 266.98	14.3%	48.48 ± 8.00
Synthetic	90	10	Water heavy sensitivity	41.34 ± 2.93	68.40 ± 4.81	73.8%	9123.56 ± 270.96	14.3%	99.98 ± 18.62

7 Decision এবং limitations

প্রথম full matrix one-slot cache policy retain করার পক্ষে evidence দিয়েছে। দ্বিতীয় slot এবং adaptive prior আপাতত defer করা হয়েছে, কারণ অধিকাংশ loaded parcel unused অবস্থায় return করেছে এবং water-heavy sensitivity cache energy overhead বাড়িয়েছে। Item prior field-calibrated probability নয়। Matrix bounded planner settings ব্যবহার করেছে এবং unrestricted optimality claim করে না।

8 Evidence এবং reproducibility

Raw metrics, decision traces, summary CSV এবং manifest Simulation results folder-এ সংরক্ষিত। V2.1 code আলাদা v2_1 namespace-এ এবং এই report-এর editable source V2.1 report folder-এ রাখা হয়েছে।